

Development of Exosome-encapsulated Paclitaxel to Overcome MDR in Cancer cells

Nanomedicine : nanotechnology, biology, and medicine

BACKGROUND

The recently emerged field of nanotechnology holds great promise for developing drug delivery systems with targeting and controlled-release characteristics for cancer treatment; there have been many new advances and innovations made in this field during the past decade ^[1]. A large proportion of chemotherapeutic drugs have low aqueous solubility, consequently requiring the use of specialized delivery vehicles (liposome, polymeric nanoparticles, *etc.*) for parenteral administration. Regrettably, many of these synthetic drug delivery vehicles cause severe side effects, including organ toxicity and/or immune response.

Another factor that considerably limits the efficacy of chemotherapeutics is multiple drug resistance (MDR), which may be intrinsic or acquired ^{[2], [3]}, and is mediated by different mechanisms, in particular, the overexpression of the drug efflux transporter P-glycoprotein (Pgp). As a result, the response rate following treatment remains very low for many types of malignancies, including, malignant gliomas, metastatic breast cancer, *etc.* ^{[2],[4]}. To date, there has been limited success in overcoming drug resistance in cancers through the use of novel small molecule chemotherapeutics ^{[5], [6]}, or nanoformulations of existing chemotherapeutics ^{[7]-[9]}. These efforts did result in improved patient outcomes; however, the non-specific inhibition of Pgp frequently increased drug toxicity due to alteration of drug elimination pathways in the liver, kidney, *etc.* ^[10].

Exosomes are membrane-derived vesicles ~40 - 200 nm in diameter ^[11] that can be found in extracellular bodily fluids (*e.g.* urine, saliva, cerebrospinal fluid) and in conditioned cell culture media ^[12]. They are formed in multivesicular bodies (MVB) inside the cells ^{[5], [12]}. Exosomes naturally function as intercellular messengers, carrying RNA and proteins ^[13]. Recently, exosomes have begun to be explored for use as drug delivery vehicles for nucleic acids ^{[14]-[19]}, gene delivery ^[20], and small molecule drugs, such as curcumin ^{[12], [21], [22]}, and doxorubicin ^[23]. It was demonstrated that exosomes have the exceptional ability to interact with cellular membranes, and deliver their payload to target cells ^{[24], [25]}. Several studies indicate that exosomes may have a specific cell tropism, according to their characteristics and origin, which can be used to target them to disease tissues and/or organs ^[26]. Furthermore, collected from patients' tissues or blood-born immunocytes, allogenic exosomes may have an immune privileged status, which allows for decreased drug clearance and immune response compared to PEGylated nanoformulations ^[27]. Thus, exosomes may function as an "invisibility cloak" for incorporated therapeutic agents, diminishing clearance by the mononuclear phagocyte system and concurrently increasing drug transport to target tissues. Thus, exosomes may comprise advantages of both synthetic nanocarriers and cell-mediated drug delivery, avoiding the rapid clearance and toxicity associated with synthetic vehicles, as well as the complexity in utilizing cell-mediated drug delivery systems in clinic. These unique features make exosomes an attractive option for use as a drug delivery vehicle for cancer treatment.

Herein, we have developed a new exosome-based formulation of PTX, a commonly used chemotherapeutic agent. The main objectives of this study were: (i) prepare and characterize a new formulation of exosomes loaded with PTX (exoPTX), (ii) assess the feasibility of using exoPTX for MDR-related anticancer therapy, and (iii) identify mechanisms underlying the effects of exoPTX in MDR cancer cells. Our results indicate that exosomes may represent a promising novel delivery platform for treatment of MDR neoplasms.

METHODS

Reagents

PTX and Doxorubicin (DOX) was purchased from LC Laboratories (Woburn, MA). Lipophilic fluorescent dyes, 1,1'-dioctadecyl-3,3',3'-tetramethylindo-carbocyanine perchlorate (DIL), and 2-decanoyl-1-(O-(11-(4,4-difluoro-5,7-dimethyl-4-bora-3a,4a-diaza-s-indacene-3-propionyl-amino)undecyl)-sn-glycero-3-phosphocholine (BODIPY-PC), were purchased from Invitrogen (Carlsbad, CA,) and Molecular Probes (Eugene, OR), respectively. Rhodamine 123 (R123), 4',6-diamidino-2-phenylindole dihydrochloride (DAPI), and Triton X-100 were obtained from Sigma-Aldrich (St. Louis, MO,). Cell culture medium and fetal bovine serum (FBS) were purchased from Gibco Life Technologies, (Grand Island, NY). Fluorescent polystyrene nanoparticles (Fluoro-Max G100) were obtained from Thermo Fisher Scientific (Waltham, MA). ExoQuick-TC™ Exosome Precipitation Solution was obtained from System Biosciences (Mountain View, CA).

Cells

RAW 264.7 macrophages, Madin-Darby canine kidney MDCK_{WT} and MDCK_{MDR1} cells were purchased from ATCC (Manassas, VA) and cultured in Dulbecco's modified Eagle's medium (DMEM) high glucose (Gibco) supplemented with 10% FBS, 1% penicillin and streptomycin at 37°C and 5% CO₂. Murine Lewis lung carcinoma cell subline (3LL-M27), a highly metastatic lung clone, was a generous gift from Dr. L. Pelletier (CHUL, Laval University, QC, Canada).

Pgp protein levels in sensitive and resistant cancer cells were determined by western blot as previously reported ([28]) using monoclonal antibodies to Pgp, C219 (Dako Corp., Carpinteria, CA; at dilution 1:100), and secondary horseradish peroxidase donkey anti-mouse IgG antibodies (Amersham Life Sciences, Cleveland, OH; at dilution 1:1500). To correct for loading differences, the Pgp levels were normalized to the constitutively expressed β -actin stained with anti- β -1-chicken integrin (Sigma Chemical Co., at dilution 1:200). Specific bands were visualized using a chemiluminescence kit (Pierce, Rockford, IL).

Characterization of Exosomes

Exosomes were harvested from the conditioned media of RAW 264.7 cells cultured in exosome-depleted media using the ExoQuick-TC™ Kit (System BioSciences; Mountain View, CA) and characterized by Nanoparticle Tracking Analysis (NTA), Dynamic Light Scattering (DLS), Atomic Force Microscopy (AFM) and Western Blot Analysis as described previously ([29]).

BODIPY-PC was used as a probe to examine the fluidic properties of exosomal membranes as described earlier ([30]). Briefly, 30 μ L exosomes with a concentration of 4×10^{11} particles/mL were mixed with 20 μ L BODIPY-PC (0.03 mg/ml) ([31]) and supplemented with 70 μ L deionized water; the mixture was incubated for 45 min at 37°C in the dark. Unbound label was removed using a Zeba™ column (Life Technologies).

Drug Loading into Exosomes

For PTX and DOX loading into exosomes, purified exosomes ($\sim 10^{11}$ exosomes) were first mixed with PTX or DOX in 1 mL PBS. Different methods of drug loading were investigated: incubation at room temperature (RT), electroporation, and sonication. *For the incubation method*, the admixture was incubated at 37°C for 1 hour with shaking. *For the electroporation method*, exosomes were mixed with PTX and added to a chilled 4 mm electroporation cuvette. The mixture was then electroporated using an Eppendorf Eporator (Eppendorf AG, Hamburg, Germany) at 1000 kV for 5 ms, and then incubated at 37°C for 30 min to allow for recovery of the exosomal membrane. *For the sonication method*, the PTX-exosome or DOX-exosome mixture was sonicated using a Model 505 Sonic Dismembrator with .25" tip with the following settings: 20% amplitude, 6 cycles of 30 s on/off for three minutes with a two minute cooling period between

each cycle. After sonication, exoPTX or exoDOX solution were incubated at 37°C for 60 min to allow for recovery of the exosomal membrane. Excess free drug was separated from exoPTX or exoDOX by size exclusion chromatography using a NAP-10 Sephadex G25 column (GE Healthcare, Buckinghamshire, UK).

The amount of PTX loaded into exosomes was measured by a high performance liquid chromatography (HPLC) method. Briefly, exoPTX (10^{10} exosomes/0.1mL) in a microcentrifuge tube was placed on a heating block set to 75°C to evaporate solvent. Then, an equal volume of acetonitrile was added and the mixture was vortexed, sonicated and then centrifuged at 13,000 rpm (Thermo Legend Micro 21) for 10 min. Following centrifugation, the supernatant was taken and filtered through a Corning Regenerated Cellulose .2 μ m syringe filter and transferred into HPLC autosampler vials. 20 μ L aliquots were injected into the HPLC system (Agilent 1200, Agilent Technologies, Palo Alto, CA). All analyses were performed using a C18 column (Supelco Nucleosil C18, 250 mm $\times$ 4.6 mm, 5 μ m, 100 Å, Sigma-Aldrich,) with a mobile phase of H₂O:acetonitrile (45:55, v/v) at a flow rate of 1 mL/min at 30°C. Absorbance was measured at 227 nm to monitor the elution of PTX.

To measure PTX release, freshly prepared exoPTX were placed in a 300K MWCO Float-A-Lyzer G2 device (Spectrum Laboratories, Houston, TX). The device was then placed in PBS under sink conditions at RT with stirring. Samples were taken at time points from inside the dialysis tube and were analyzed by HPLC as described above. The amount of PTX released from exoPTX was expressed as a percentage of total PTX and plotted as a function of time.

Accumulation of Exosomes and Exosome-incorporated PTX in Cancer Cells

To quantify the amount of exosomes taken up by cells, exosomes were stained with a lipophilic fluorescent dye, DIL as described previously ([29]). Then, DIL-labeled exosomes, or fluorescently-labeled liposomes, or polystyrene nanoparticles (NPs, Fluoro-Max G100, Thermo Fisher Scientific), were added in equal numbers ($\sim 10^8$ particles/well) and incubated with 3LL-M27 cells at 37°C and 5% CO₂ for various times. After each time point, the media was removed and cells were washed 3x with PBS and fixed by incubating with Formal-Fixx (Thermo Fisher Scientific), and examined by confocal microscopy or using a Shimadzu RF5000 fluorescent spectrophotometer. In case of exoPTX or Taxol, drugs were added in equimolar amounts to the MDCK_{WT}, or MDCK_{MDR1} cells and incubated for 72h. The cell suspension was then lysed and analyzed for PTX content by HPLC as described above.

In vitro Cytotoxicity Assay

The *in vitro* antitumor efficacy of exoPTX was assessed using a standard MTT (3-(4,5-dimethyl-2-thiazolyl)-2,5-diphenyl-2-H-tetrazoliumbromide) assay with three cancer cell lines, and compared to Taxol as described earlier ([32]). Briefly, cancer cells (MDCKMDR1, MDCKwt, and 3LL-M27) were seeded at 5,000 cells/well in 100 μ L of media in 96-well plates overnight. Various concentrations of exosomes isolated from macrophages conditioned media and loaded with PTX by sonication, or empty sonicated exosomes, or Taxol, or PTX were added to cancer cells for 48 hours at 37°C, 5% CO₂. Following the incubation, the cells were washed and incubated with MTT reagent as described in ([33]). The cytotoxic activity of PTX was then evaluated using a standard MTT assay ([34]). The absorbance at 570nm was measured using a Shimadzu RF5000 fluorescent spectrophotometer. The survival values were determined in relation to control cells cultured in drug-free media. All experiments were repeated at least three times. SEM values were less than 10%.

Production of a Lentiviral Vector (LV) and Transduction of LLC cells

Lentiviral vector encoding a fusion between the optical reporter mCherry (GBM8FlmC, red) and firefly luciferase (FLuc) were created by PCR amplification of the cDNA sequences for mCherry and FLuc from pEmCherry (Clontech) and pcDNA-Luciferase (Addgene) with restriction enzyme sequences that were engineered into the primers. To create the final constructs, mCherry was digested with BamHI/EcoV and FLuc was digested with EcoV/XhoI. The digested

fragments were ligated into the BamHI/XhoI digested pTK402 LV transfer vector (a kind gift from Dr. Tal Kafri, The University of North Carolina at Chapel Hill). LV-mCherryFLuc viral vectors were packaged in 3LL-M27 cells by transient transfection using the psPAX2 and pMD2.G (Addgene) packaging plasmids and following previously described protocols ([35]).

To utilize bioluminescence and fluorescence imaging, 3LL-M27 cells were transduced with lentiviral vectors encoding an mCherry and Renilla luciferase (mC-RL) fusion protein. The viral construct also encoded for a puromycin resistance gene downstream of mCherry which was introduced to enable for selection of nearly 100% positively transduced cells. A robust expression of both the fluorescent and bioluminescent markers was observed, and no difference in proliferation was detected between modified and unmodified cells. These cells (8FlmC-FLuc-3LL-M27) were used for biodistribution and therapeutic efficacy studies.

Biodistribution of Exosomes in Mice with Pulmonary Metastases

The experiments were performed with female C57BL/6 mice (Charles River Laboratories, Durham, NC) eight weeks of age in strict accordance with the recommendations in the Guide for the Care and Use of Laboratory Animals of the National Institutes of Health. The animals were kept five per cage with an air filter cover under light- (12-hours light/dark cycle) and in a temperature-controlled ($22 \pm 1^\circ \text{C}$) environment. All manipulations with the animals were performed under a sterilized laminar hood. Food and water were given *ad libitum*.

C57BL/6 mice ($n = 4$) were injected intra tail vein (*i.v.*) with 8FlmC-FLuc-3LL-M27 cells (5×10^6 cells/mouse in 100 μL saline) and tumor lung metastases were allowed to establish for 10-12 days. Twelve days following cancer cells *i.v.* injection, DID-labeled exosomes isolated from autologous macrophages were administered intranasally (*i.n.*, 10^7 particles/ $10 \mu\text{l} \times 2$) to mice with lung metastases. Four hours later, mice were sacrificed, perfused, lungs were extracted and sectioned on a microtome at a thickness of 20 μm ; nuclei were stained with DAPI (300 mM, 5 min). The images of lung sections were examined by a confocal fluorescence microscopic system ACAS-570 and corresponding filter set, and processed using ImageJ software.

In another experiment, mice with established GBM8FlmC-metastases were injected *i.n.* with non-labeled exosomes loaded with Dox by sonication as described above (10^7 particles/ $10 \mu\text{L} \times 2$). Four hours later mice were sacrificed, perfused; lungs were extracted, sectioned, and co-localization of Dox with pulmonary metastases was visualized by confocal microscopy.

Therapeutic Efficacy of exoPTX against Pulmonary Metastases

The antineoplastic effects of exoPTX were evaluated in a mouse model of pulmonary metastases. For this purpose, C57BL/6 mice were *i.v.* injected with 8FlmC-FLuc-3LL-M27 cancer cells (5×10^6 cells/ $100 \mu\text{L}$ / mouse). Forty eight hours later, mice were treated *i.n.* with exoPTX (10^7 particles/ $10 \mu\text{l} \times 2$), or Taxol (50 mg/kg/mouse), or saline as a control ($n = 7$) every other day with a total of seven treatments. Tumor progression was monitored by luminescence using the IVIS system as described in ([36]). The animals were imaged at various time points (1 – 22 days) post-treatment as described ([36]). The chemoluminescent signal was quantified by Living Image® 2.50 software. To assess the amount of cancer metastases at day 22, mice were sacrificed, perfused, and lung slides obtained on microtome (Thermo Scientific) were examined by confocal microscopy.

Statistical Analysis

For the all experiments, data are presented as the mean $\pm$ S.E.M. Tests for significant differences between the groups were performed using a t-test or one-way ANOVA with multiple comparisons (Fisher's pairwise comparisons) using

GraphPad Prism 5.0 (GraphPad software, San Diego, CA). A minimum p value of 0.05 was chosen as the significance level.

RESULTS

Manufacture and characterization of exosomal formulations of PTX (exoPTX)

Exosomes collected from the conditioned media of RAW 264.7 macrophages were characterized by size, charge, protein content, and morphology ([Fig. 1A, B, and D]). Exosomes showed elevated expression of exosome-associated proteins (Alix, TSG101, and Flotillin) as compared to cell lysate, which displayed greater levels of β -actin ([Fig. 1B]). Naïve empty exosomes had a narrow size distribution, with an average particle diameter of 110.4 ± 4.2 nm and 70.8 ± 2.8 nm as revealed by NTA and DLS, respectively ([Fig. 1A]); and a round morphology as shown by AFM imaging ([Fig. 1D]).

PTX was incorporated into exosomes using three methods: a) incubation at room temperature (RT), b) electroporation, and c) mild sonication. The obtained exoPTX formulations were purified from the non-incorporated drug by size-exclusion chromatography and analyzed by HPLC to determine the loading capacity (LC). The typical HPLC profiles for PTX extracted from exosomes (B) and PTX standards (A) are shown on [Supplemental Fig. 1]. The amount of PTX loaded into exosomes increased as follows: incubation at RT < electroporation << sonication ([Fig. 1A]). Interestingly, DLS studies revealed that the size of exoPTX nanoformulations increased similarly, with the smaller being exoPTX nanoparticles obtained by electroporation or incubation at RT, and the larger being exosomes loaded with PTX by sonication ([Fig. 1A]). These data were confirmed by NTA analysis. Exosomes sonicated in the absence of PTX were even larger than those sonicated with PTX ([Fig. 1A]). We hypothesized this may be due to the stabilization of exosomal membranes by the incorporated drug. We suggested that a reorganization of exosomal membranes under sonication may enable PTX diffusion across relatively tight lipid bilayers. Indeed, fluorescence polarization measurements revealed significant decreases (more than two times) in membrane microviscosity upon sonication ([Fig. 1C]). To address a concern about possible loosing of exosome-bound proteins, we examined the levels of Alix, TSG101, and Flotillin in exosomes before and after sonication using western blot technique ([Fig. 1B]). The data indicate that the mild sonication utilized for PTX loading with six cycles, and intermediate time out for cooling down and restoration, did not significantly affect the protein content of exosomes. It is known that the anionic phospholipid phosphatidylserine is abundant on cell membranes and contributes to the surface charge of individual cellular membranes. To this end, all loading procedures did not significantly alter the zeta potential of the nanocarriers ([Fig. 1A]), suggesting that there were also no major alterations of the lipid content of exosomal membranes. Finally, a complete restoration of membrane microviscosity was observed after a one hour incubation at 37C following sonication procedure ([Fig. 1C]). Retention of shape and round morphology of exosomes ([Fig. 1D]) confirmed this hypothesis.

Next, exoPTX showed burst release within the first three hours, and then displayed a sustained release profile thereafter ([Fig. 1E]). The high stability of exosomes in an aqueous solution was demonstrated at three temperatures: 4°C, RT, and 37°C over a period of one month ([Supplemental Fig. 2]). Overall, the mild sonication procedure provided the highest amount of drug loading; the obtained LC of $28.29 \pm 1.38\%$ ([Fig. 1A]) was much higher than the LC of commercially available formulations of PTX, Taxol (~1% LC), or Abraxane (~10% LC). Therefore, exoPTX obtained by sonication was selected for further experiments.

Accumulation and therapeutic efficacy of exoPTX in target cancer cells in vitro

The ability to deliver the drug payload into target cells was studied with fluorescently-labeled exosomes in 3LL-M27 cells, and compared to the commonly used nanocarriers, liposomes and polystyrene nanoparticles (NPs) (^[37]) with the

same size and level of fluorescence (Fig. 2I). Liposomes were prepared by a reverse phase evaporation method as described previously [29]. Confocal images revealed a profound accumulation of exosomes in cancer cells and limited uptake of liposomes and NPs (Fig. 2A). This result was further confirmed and quantitated in accumulation studies (Fig. 2B). Exosomes were taken up about 30 times better than the synthetic nanoparticles, suggesting that PTX loaded into exosomes can be efficiently delivered to cancer cells in therapeutically sufficient quantities. These results clearly show the advantages of exosome-based drug delivery systems over common synthetic nanocarriers and confirmed our previous report regarding the profound accumulation of exosomes in neuronal PC12 cells [29].

The anticancer effects of exoPTX were evaluated in a resistant MDR cells expressing the drug efflux transporter, Pgp (MDCK_{MDR1}), and their sensitive counterparts (MDCK_{WT}). The loading of PTX into exosomes significantly increased drug cytotoxicity as compared to PTX alone, or Taxol in both sensitive MDCK_{WT} and resistant MDCK_{MDR1} cancer cells (Table 1). These results are consistent with earlier reports regarding increased cytotoxicity of another anticancer agent, Dox in cancer cells [23]. The most intriguing observation was made, when the effects of various PTX formulations were compared in sensitive and resistant cancer cells. For this purpose, the increased cytotoxicity of the drug was expressed in the form of a “Resistance Reversion Index” (*RRI*), i.e. ratio of IC_{50} of PTX alone, and in nanoformulation (e.g. $IC_{50,PTX}/IC_{50,exoPTX}$, or $IC_{50,PTX}/IC_{50,taxol}$). Both PTX formulations caused significant sensitization of MDR cells with respect to PTX (Table 1). In particular, *RRI* for exoPTX in MDCK_{MDR1} and MDCK_{WT} was 53.33 and 18.38, respectively. In contrast, *RRI* for Taxol in both resistant and sensitive cancer cells was *c.a.* 6 (Table 1). Noteworthy, empty sonicated exosomes did not show any cytotoxicity in all studied cell lines (Supplemental Fig. 3). Thus, the increase in PTX cytotoxicity afforded by exoPTX was greater in Pgp-overexpressing cells than their sensitive counterparts (Table 1).

Mechanistic studies of exoPTX cytotoxic effects

We hypothesized that exoPTX may alter drug intracellular trafficking and bypass the drug efflux system more efficiently than Taxol (in particular, exoPTX may facilitate endosomal release of PTX from exosomes in cancer cells). To prove this hypothesis, we examined the accumulation levels of a fluorescent probe and Pgp substrate, Dox, incorporated into exosomes (exoDox) in MDCK_{MDR1} and MDCK_{WT} cells. First, elevated Pgp expression levels in MDCK_{MDR1} cells, and low, if any, Pgp levels in MDCK_{WT} cells were confirmed by western blot (Fig. 3A). Next, the uptake of free Dox and exosome-incorporated drug, exoDox, was compared in the presence/absence of a Pgp inhibitor, verapamil. As expected, the incorporation of Dox into exosomes significantly increased drug accumulation levels in both sensitive and resistant cancer cells (Fig. 3B). Inhibition of Pgp-mediated drug efflux by verapamil increased accumulation of free Dox in resistant MDCK_{MDR1} cells, but did not alter drug accumulation in their sensitive counterparts. Remarkably, verapamil treatment did not affect exoDox accumulation in resistant MDCK_{MDR1} cells, indicating that drug incorporation into exosomes allowed it to bypass this resistance mechanism (Fig. 3B).

We demonstrated earlier that the incorporation of Pgp substrates, such as R123 or Dox, into block-copolymer-based nanocarriers, i.e. Pluronic micelles, increased drug accumulation in resistant cancer cells due to the inhibition of Pgp efflux transporter by Pluronic macromolecules incorporated into the membranes of resistant cancer cells [28], [38]–[40]. To exclude the possibility that exosomes may inhibit Pgp-mediated efflux by their fusion with cellular membranes, accumulation of R123 in both resistant and sensitive MDCK cancer cells was assessed. R123 does not incorporate into exosomes upon incubation at RT, as was confirmed in our preliminary studies (Supplemental Fig. 4). For this purpose, MDCK_{WT} and MDCK_{MDR1} cell monolayers were pretreated with a Pgp inhibitor, verapamil (positive control), or empty

exosomes, or media (negative control), and then were treated with R123 solutions for two hours ([Supplemental Fig. 5]). R123 accumulation levels in resistant MDCK_{MDR1} cells were increased almost five times in verapamil pre-treated cells. In contrast, treatment with empty exosomes did not affect R123 accumulation in MDCK_{MDR1} cells ([Supplemental Fig. 5]). As expected, neither treatment with verapamil, nor with empty exosomes, altered R123 accumulation levels in sensitive MDCK_{WT} cells. This indicates that exosomes themselves do not appear to have any inhibitory effect on Pgp-mediated efflux; they allow incorporated drugs to bypass the Pgp efflux protein perhaps, through endocytosis-mediated transport and/or fusion with plasma membranes.

Co-localization of Airway-delivered Exosomes with Pulmonary Metastases in Lewis Lung Carcinoma (LLC) mouse model

To establish an *in vivo* model of pulmonary metastases, C57BL/6 mice were injected intra-tail vein (5×10^6 cells/100 μ L) with 3LL-M27 cells. Important, this model is particularly relevant to the present investigation, as it was demonstrated that 3LL-M27 tumor cells have high expression levels of the MDR1 gene and Pgp expression *in vivo* ([³²]). Twenty days later, mice were sacrificed, perfused, and lungs were isolated, sectioned, and stained with Hematoxylin and Eosin (H&E). Multiple metastases were detected in whole lungs ([Supplemental Fig. 6A]). Histological evaluations revealed that the structure of alveoli in tumor-bearing lungs was disrupted by tumor cells ([Supplemental Fig. 6B]). Next, mice were injected with 8FlmC-FLuc-3LL-M27 (red, [Fig. 4]) intra-tail vein as described in Materials and Methods section. 22 days later, autologous exosomes stained with a fluorescent dye, DiD (green), were *i.n.* administered to mice with pulmonary metastases. Four hours later, mice were sacrificed, perfused; lungs were sectioned on microtome and examined by confocal microscopy. Nuclei were stained with DAPI (blue, [Fig. 4]). Confocal images revealed $97.9 \pm 2.0\%$ of exosomes were co-localized with lung metastases ([Fig. 4]), indicating efficient targeting of exoPTX *in vivo*. A similar experiment was performed with exoDox formulation in order to visualize drug delivery to pulmonary metastases ([Supplemental Fig. 7]). Non-labeled exosomes loaded with Dox (green, [Supplemental Fig. 7B]) were *i.n.* administered to mice with established 8FlmC-FLuc-3LL-M27 metastases (red, [Supplemental Fig. 7A]). Confocal images revealed a substantial amount of DOX in the lungs co-localized with cancer cells (yellow, [Supplemental Fig. 7C]). These results indicate that airway-administered exosomes reached pulmonary metastases and delivered their drug payload to target cancer cells.

PTX-loaded exosomes produce strong antineoplastic effect in mice with lung metastases

To provide insight into the potential of exosome-based therapeutic delivery, the antineoplastic effects of exoPTX were evaluated in an LLC mouse model. For this purpose, C57BL/6 mice were *i.v.* injected with 8FlmC-FLuc-3LL-M27 cells as described above. 48 hours later, mice were *i.n.* administered exoPTX (10^7 particles/10 μ l $\times$ 2), or Taxol, or saline as a control every other day totally, seven times. The progression of pulmonary metastases in treated mice was monitored using IVIS by observing the luminescence of transduced cancer cells in living animals ([Fig. 5A]). Representative images of dorsal planes of the injected animals at day 22 are shown on ([Fig. 5A]). A significant ($p < 0.05$) inhibition of metastases growth by exoPTX treatment was demonstrated ([Fig. 5C]). Taxol treatment was shown to inhibit metastases growth as compared to non-treated controls (saline), although to a lesser extent than exoPTX treatment. At the end point of the experiment (day 22), the lung sections were visualized using confocal microscopy ([Fig. 5B]). A marked number of fluorescent transduced cancer cells (red) were detected in the lungs of animals treated with Taxol ([Fig. 5C]), while only a few cancer cells were observed in the lungs of exoPTX treated animals. Noteworthy, sonicated empty exosomes showed no significant inhibition on pulmonary metastases growth ([Fig. 5A, B]). This confirms the superior antineoplastic efficacy of exoPTX as compared to Taxol.

DISCUSSION

Exosomal carriers can provide advantages of both cell-based drug delivery and nanotechnology for efficient drug transport capable of overcoming various biological barriers. However, several limitations need to be addressed before their use in the clinic. One of the difficulties is the efficient loading exosomes with a therapeutic agent without significant changes in the structure and content of exosomal membranes. In the present study, we utilized various methods for PTX incorporation into exosomes: incubation at RT, electroporation, and mild sonication. The mild sonication of exosomes in the presence of PTX provided the greatest loading capacity. PTX, a highly hydrophobic compound is likely to be incorporated into the hydrophobic inner region of the lipid bilayers of exosomes. We hypothesized that the high rigidity of exosomal membranes may be decreased upon sonication and would thus allow for PTX incorporation into lipid bilayers resulting in a high loading capacity. This hypothesis was confirmed by significant decreases in microviscosity of exosomal membranes upon sonication. Nevertheless, we do not exclude a possibility that a considerable amount of PTX may also be adhered to the surface of exosomes that may account for the burst release from exoPTX observed in the first 3-4 hours. It is worth noting that ~30% of loaded drug was still associated with exosomes after one week in an aqueous solution. Importantly, drug located in the inner bilayer of exosomes may also be available for use: as the exosomal membrane fuses with the cell or endosomal membrane, its intraluminal cargo may be released into the cytosol of a target cell. Next, the aggregation stability of exoPTX formulations is imperative for their use in clinic. We report here that the obtained exoPTX formulation was stable at various conditions for over a month, which confirms previous reports about the long-term stability of exosomes (^[41]). In addition, exosomes may be lyophilized and reconstituted, while retaining their morphology and other characteristics (^[29]). This provides a clinical link for exosome-based drug formulations, suggesting that multiple lots of exoPTX may be prepared and stored prior to treatment.

Exosomes possess an extraordinary ability to interact with and accumulate in target cancer cells. The obtained data indicates exosomes are taken up in considerably greater numbers than liposomes or polystyrene NPs. In addition, the incorporation of PTX into exosomes may not only increase its solubility, but also allow for overcoming of Pgp-mediated drug efflux. We demonstrated here that incorporation of a Pgp substrate, Dox, into exosomes significantly increased drug accumulation in MDR cells as compared to free DOX, or even to Dox in the presence of a Pgp-inhibitor, verapamil. Next, the increase in cytotoxicity afforded by the exosomal formulation of PTX was considerably greater in resistant cells ($RRI > 53.33$) than sensitive cells ($RRI = 18.35$), while Taxol showed almost no difference in resistant ($RRI > 5.85$) vs. sensitive cancer cells ($RRI = 6.17$). This effect may be attributed to the difference in route of internalization of exoPTX, as compared to Taxol. Exosomes and micelles, such as those found in Taxol, are taken up by endocytosis, but exosomes have superior uptake due to the presence of adhesion proteins, tetraspanins, integrins, immunoglobulins, proteoglycans, and lectins (^[42]), which are not found on artificial nanoparticles. Furthermore, exosomes consist of cellular membranes that may fuse with the plasma and/or endocytic membranes and deliver their cargo, bypassing Pgp-mediated efflux. Noteworthy, exosomes themselves did not inhibit Pgp, as the pre-treatment with empty exosomes did not increase accumulation of the Pgp substrate, R123, in resistant cancer cells.

Interestingly, it was suggested that the MDR efflux transporters are likely contribute to the production of drug-loaded exosomes during their biogenesis in resistant cancer cells (^[43]). In addition, Pgp may be also involved in the increased drug sequestration in lysosomes and MVB (^[44]). Thus, Pgp associated with the endosomal membrane excretes the internalized drug into the endosomal lumen, where newly formed cancer exosomes are literally incubated with the drug and become “drug-loaded” before being released from the cell. The same effect was reported with PTX in Pgp-overexpressing bone marrow mesenchymal stromal cells (SR4987) (^[45]). We hypothesized that exoPTX accumulated in the MDR cancer cells may bypass not only efflux by Pgp transporter located on plasma membrane, but also avoid

accumulation in lysosomes and MVB of cancer cells, and therefore, reduce drug elimination and increase its therapeutic efficacy in resistant tumors. The investigations regarding this hypothesis are underway in our laboratory.

Finally, the therapeutic efficacy of exoPTX formulation against pulmonary metastases was demonstrated in an LLC mouse model. Intriguingly, airway-delivered exosomes showed near complete co-localization with cancer metastases in this model. The results were confirmed by the significant co-localization of Dox incorporated into exosomes with cancer cells. We speculated that macrophage-released exosomes are likely to have specific proteins on their surface, which might allow for their preferential accumulation in cancer cells. Furthermore, it is known that exosome-mediated cell-to-cell communication is key in the battle between cancer and the immune system (^[46]). Thus, Parolini *et al.* (^[47]) showed that exosome fusion with target cells occurs more efficiently under acidic conditions, implying that exosomes may be taken up preferentially by tumors (which have an acidic microenvironment) rather than the surrounding healthy tissue. Our results show that exoPTX demonstrated superior inhibition of pulmonary metastases growth in LLC mouse model. All three mechanisms mentioned here are likely to have significant impact on exoPTX anticancer activity, *i.e.*: (i) preferential accumulation in cancer cells, (ii) efficient delivery of incorporated cargo into target cancer cells, and (iii) bypassing Pgp-mediated drug efflux in resistant cancer cells. Indeed, further investigations are necessary to uncover this mechanism.

Supplementary Material